



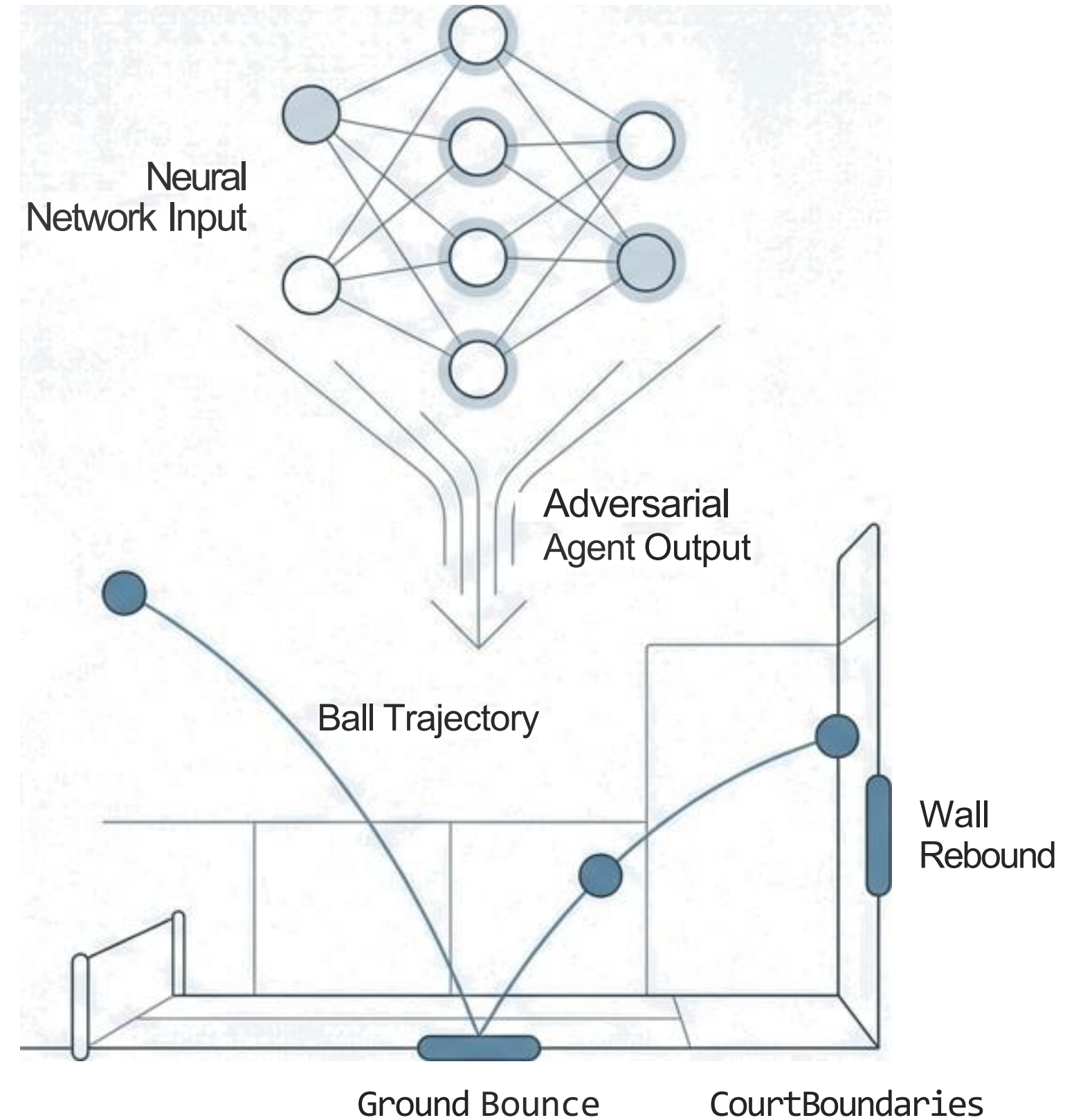
THE DIGITAL ATHLETE

Multi-Agent Reinforcement Learning: a Simplified Padel

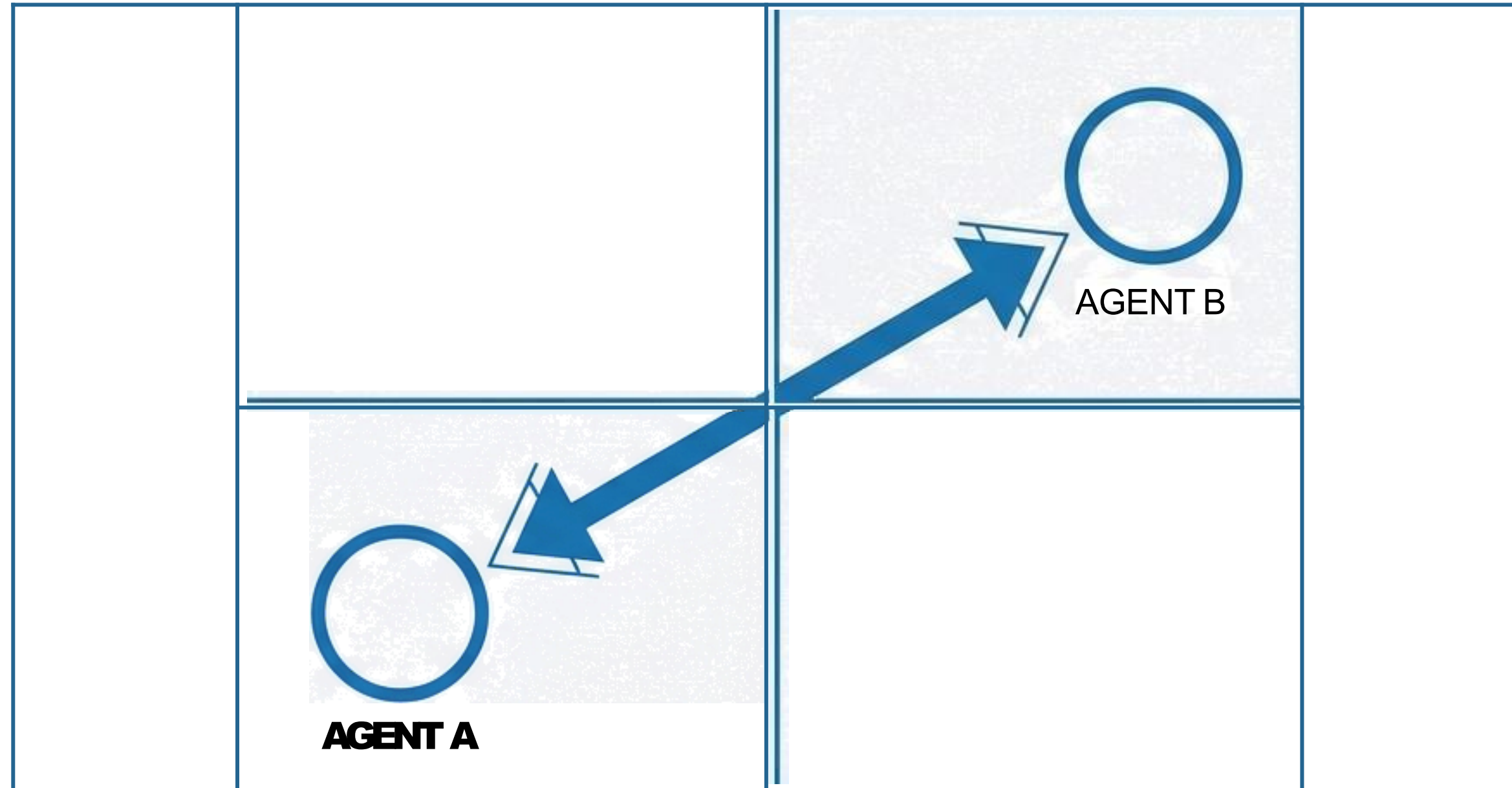
Giulia Raffaella Vitale

The Physics of Adversarial Play

Why Padel? A domain characterized by its dynamic and physically rich interactions. The simulation must master ball-ground interactions, valid/invalid bounces, and double-bounce rules.



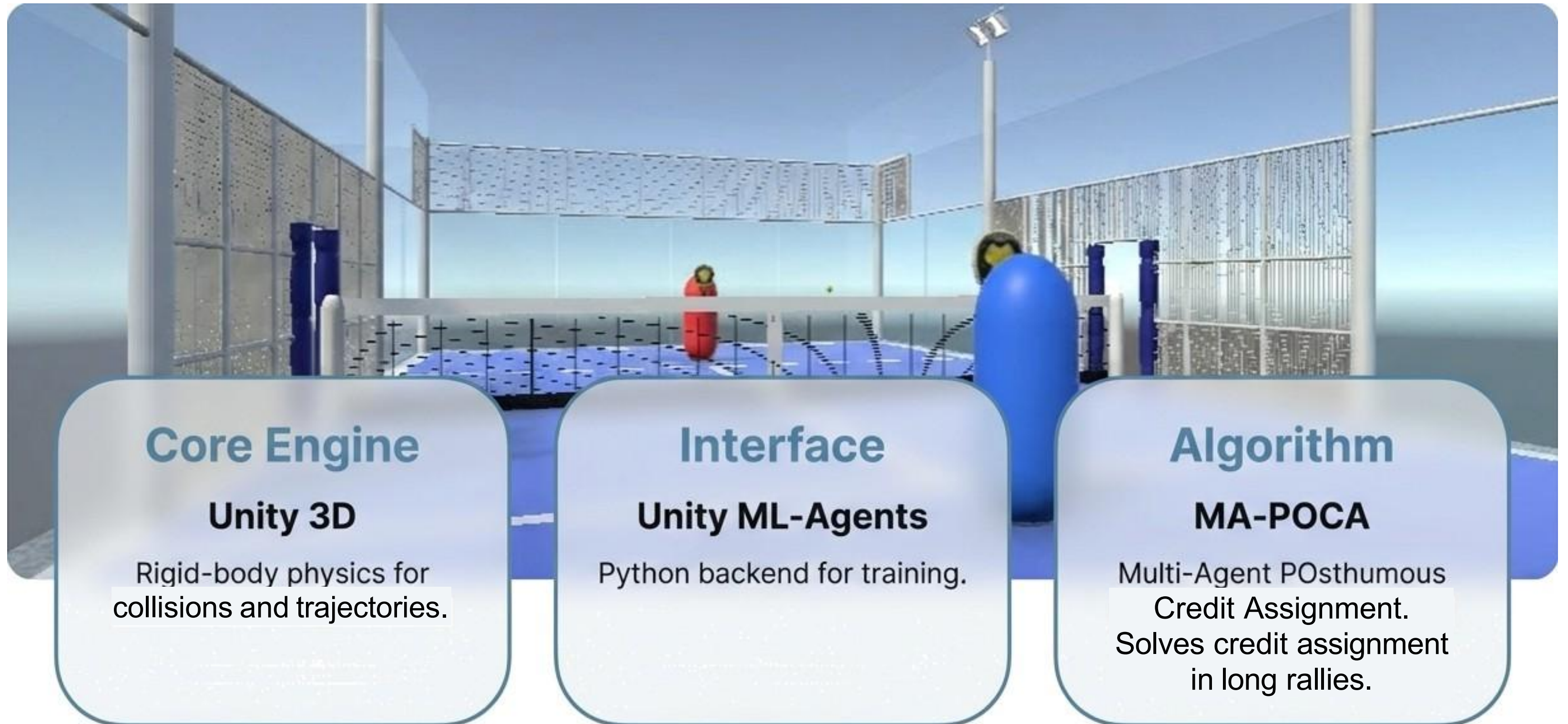
SIMPLIFYING THE BATTLEFIELD



THE STRATEGY:

- » Reduction from 2v2 to 1v1
- » Removal of Team Dynamics
- » Goal: Pure Strategic Discovery in an 'Adversarial Logic Sandbox '

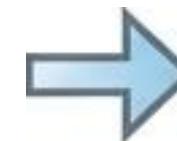
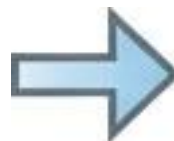
Simulation Architecture



The Markov Decision Process

Agent State

- Agent Position & Velocity
- Relative Ball Position & Velocity
- Serving Flag
- Opponent Position & Velocity



Action Space A

- Movement along X-axis
- Movement along Z-axis
- Racket swing activation

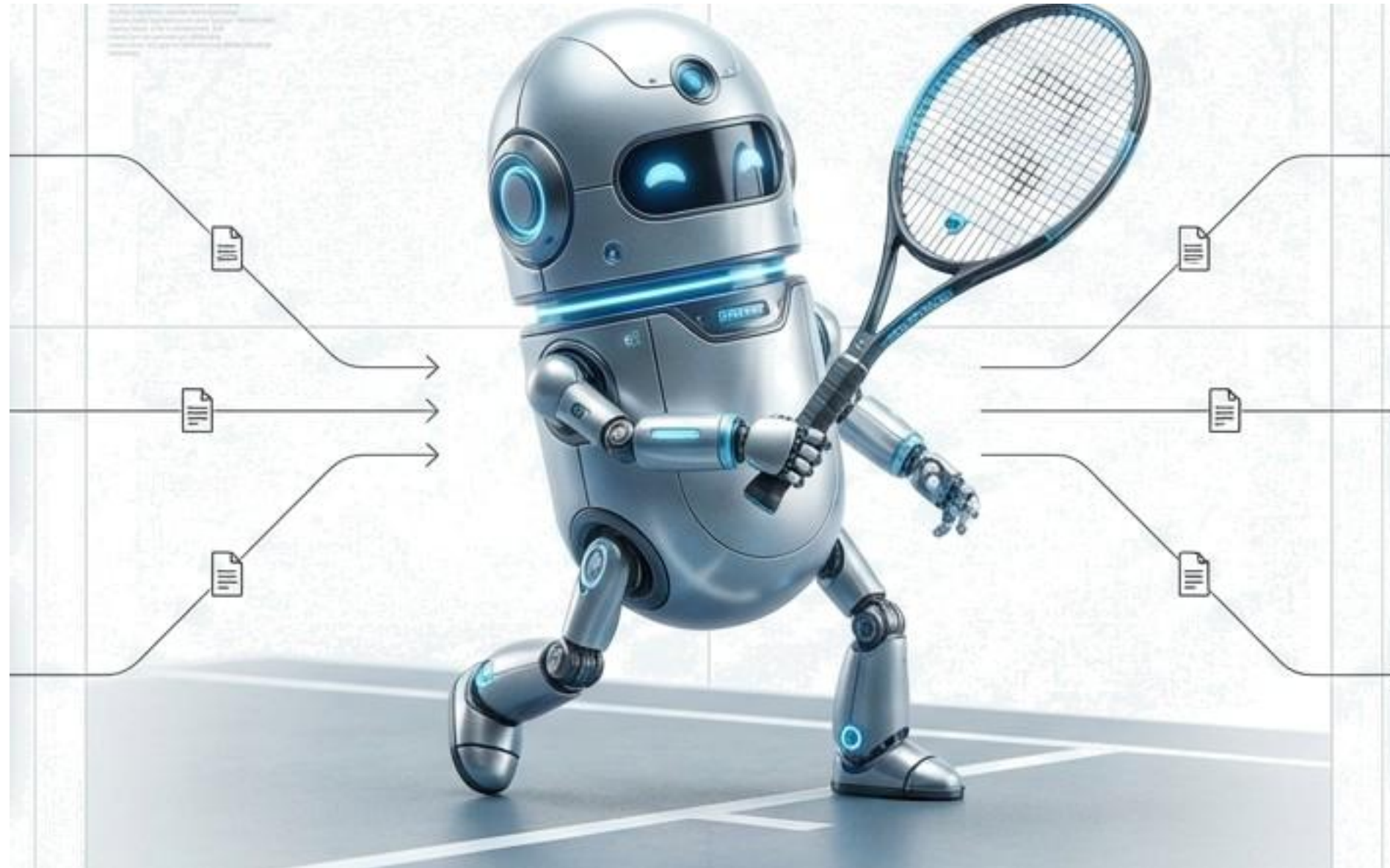
Building the Agent

INPUTS / SENSORS

Self Position
(x, y, z)

Ball Relative
Position (vector)

Opponent State



OUTPUTS / MUSCLES

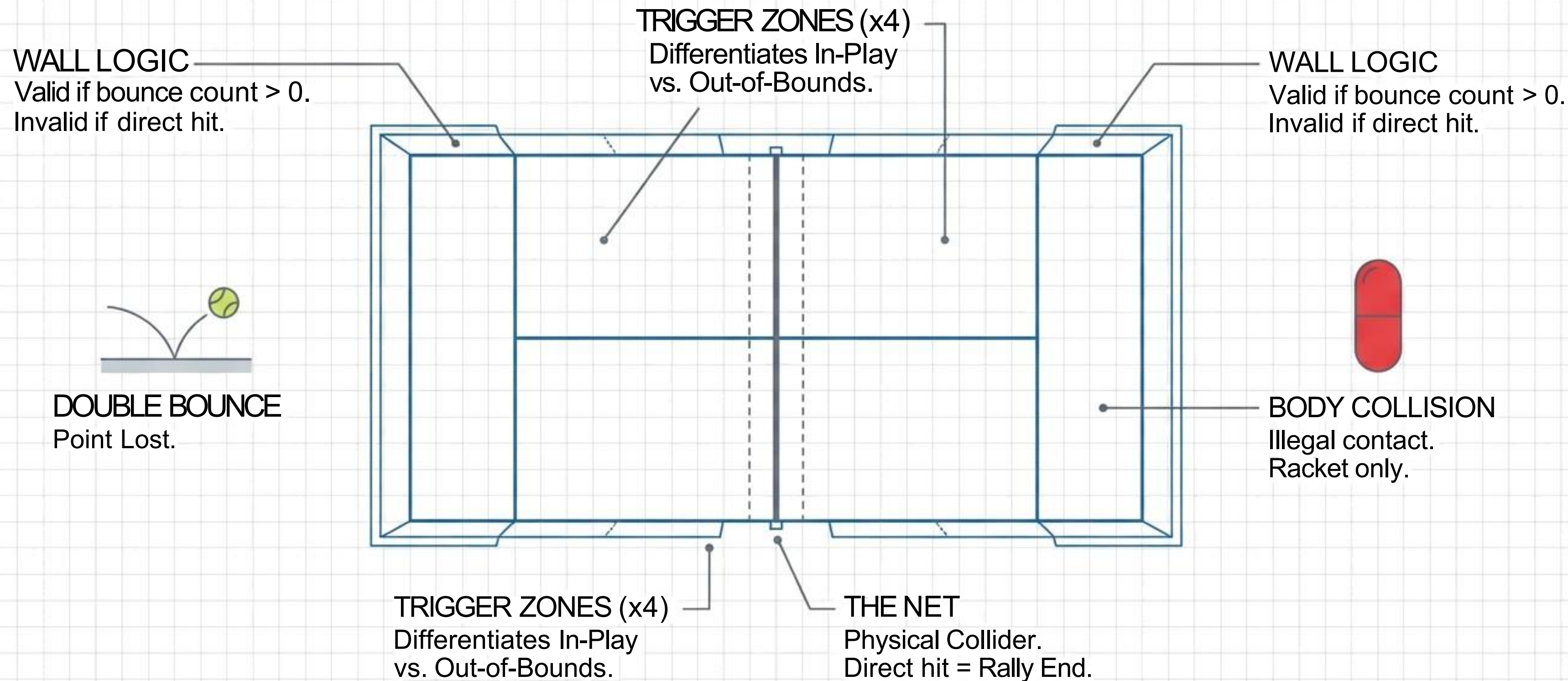
Force X
(Sideways)

Force Z
(Forward/Back)

Swing Rotation
(Torque)

Transition Function = Unity Rigid-Body Physics Engine

DEFINING THE WORLD BOUNDARIES



Shaping Behavior via Rewards

Positive Incentives (+)

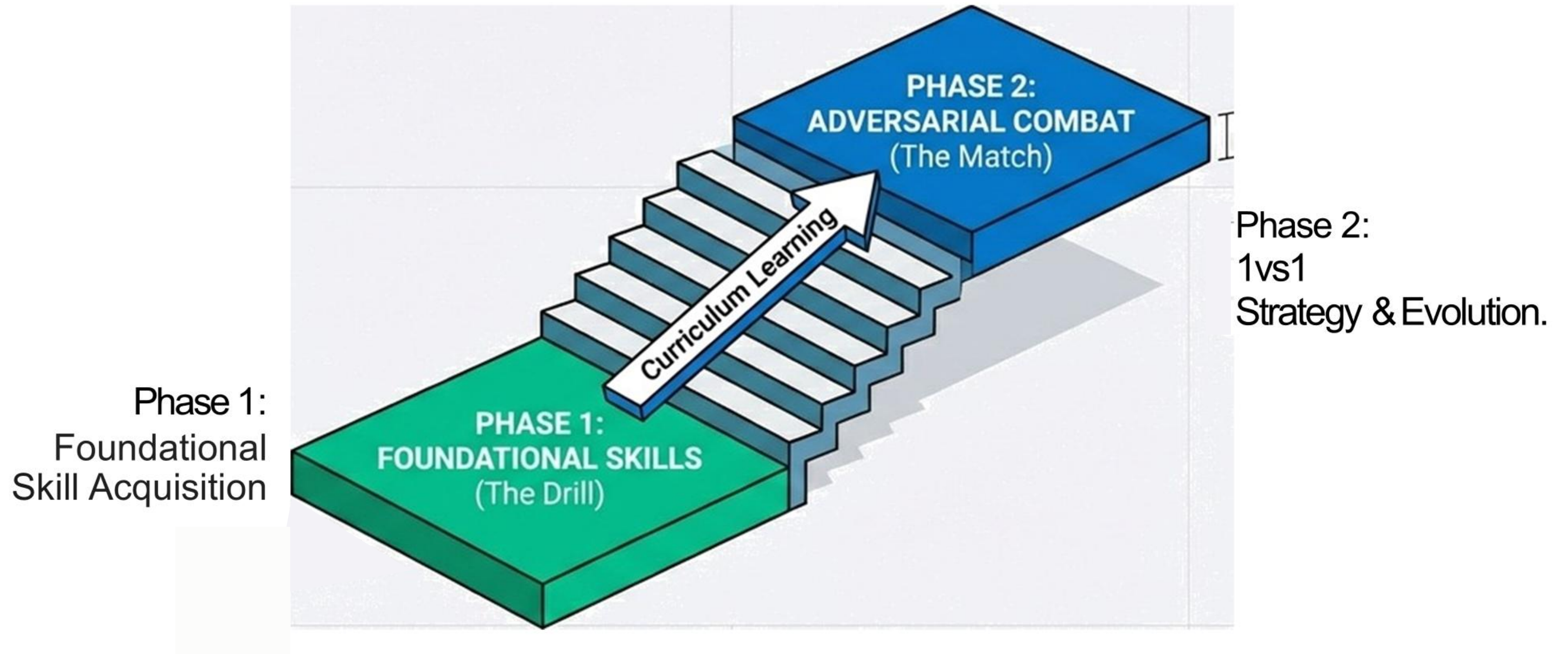
- Outcome (Winning the rally)
- Timing (Valid height window)
- Movement (Reducing distance to ball)
- Racket Hit

Negative Penalties (-)

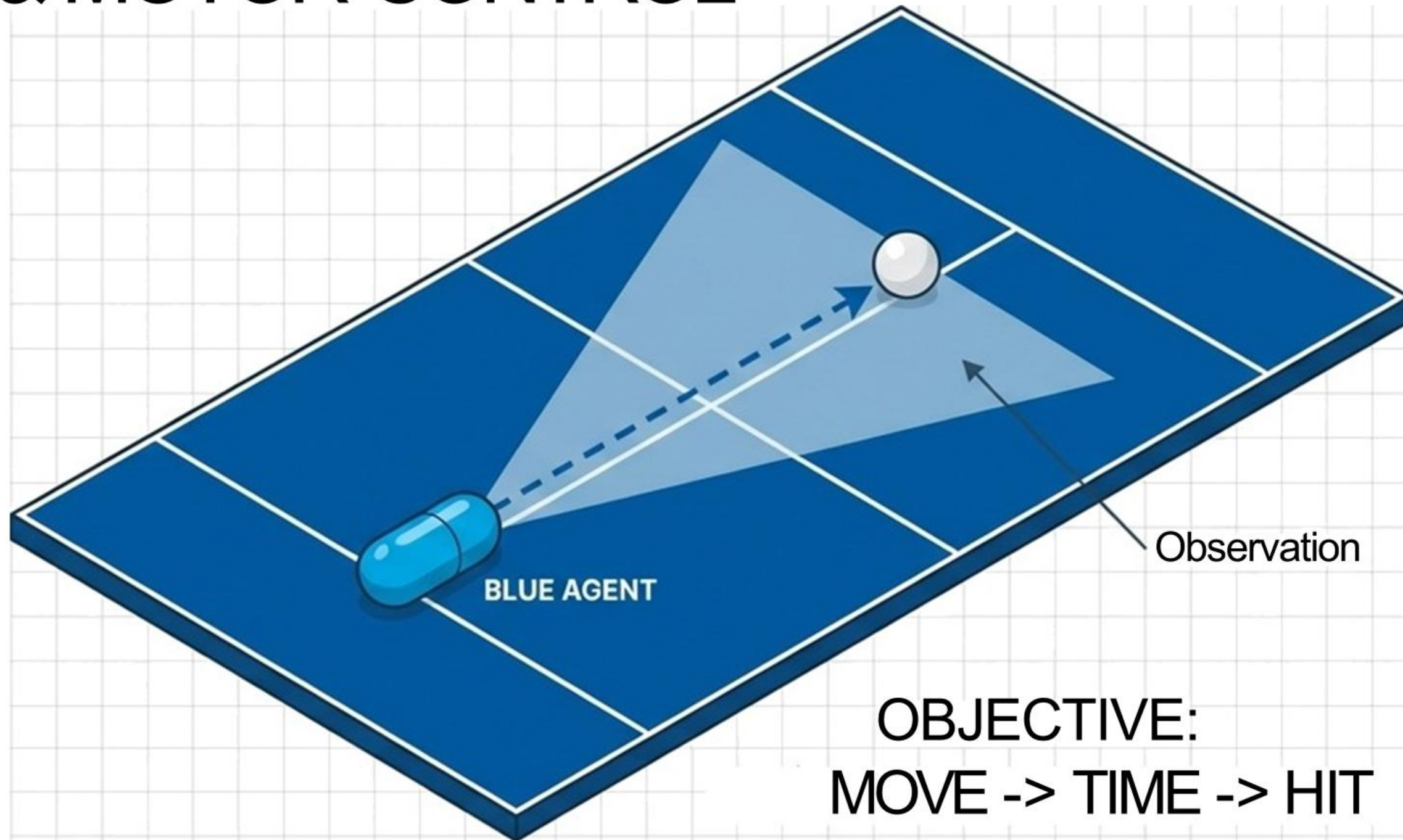
- Collisions (Net, Wall, Body)
- Violations (Out-of-bounds, Double bounce)
- Inactivity (Zero velocity)
- Losing a rally



YOU CANNOT RUN BEFORE YOU WALK



PHASE 1: SOLO DRILLS & MOTOR CONTROL



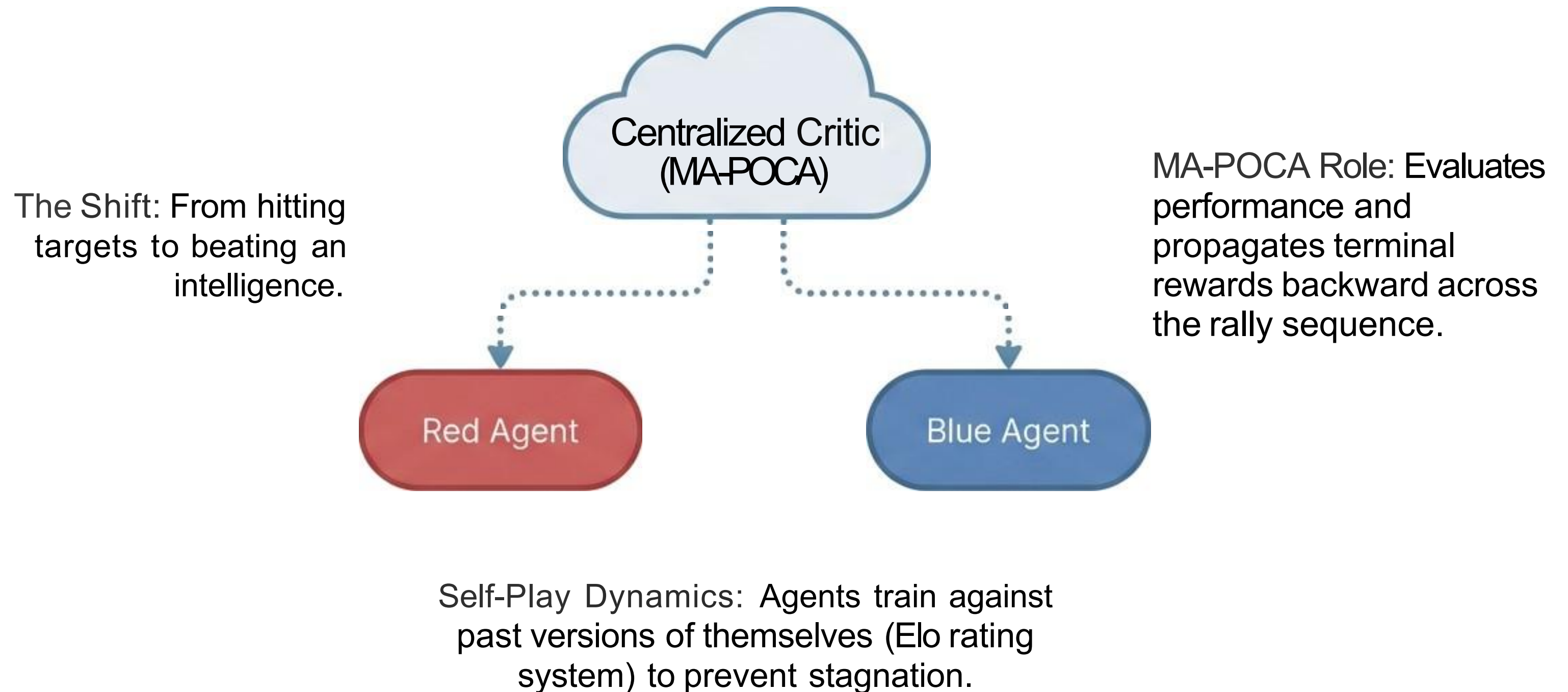
SCRIPT:
Player1step.cs

SETUP:
Single Agent, No Opponent.

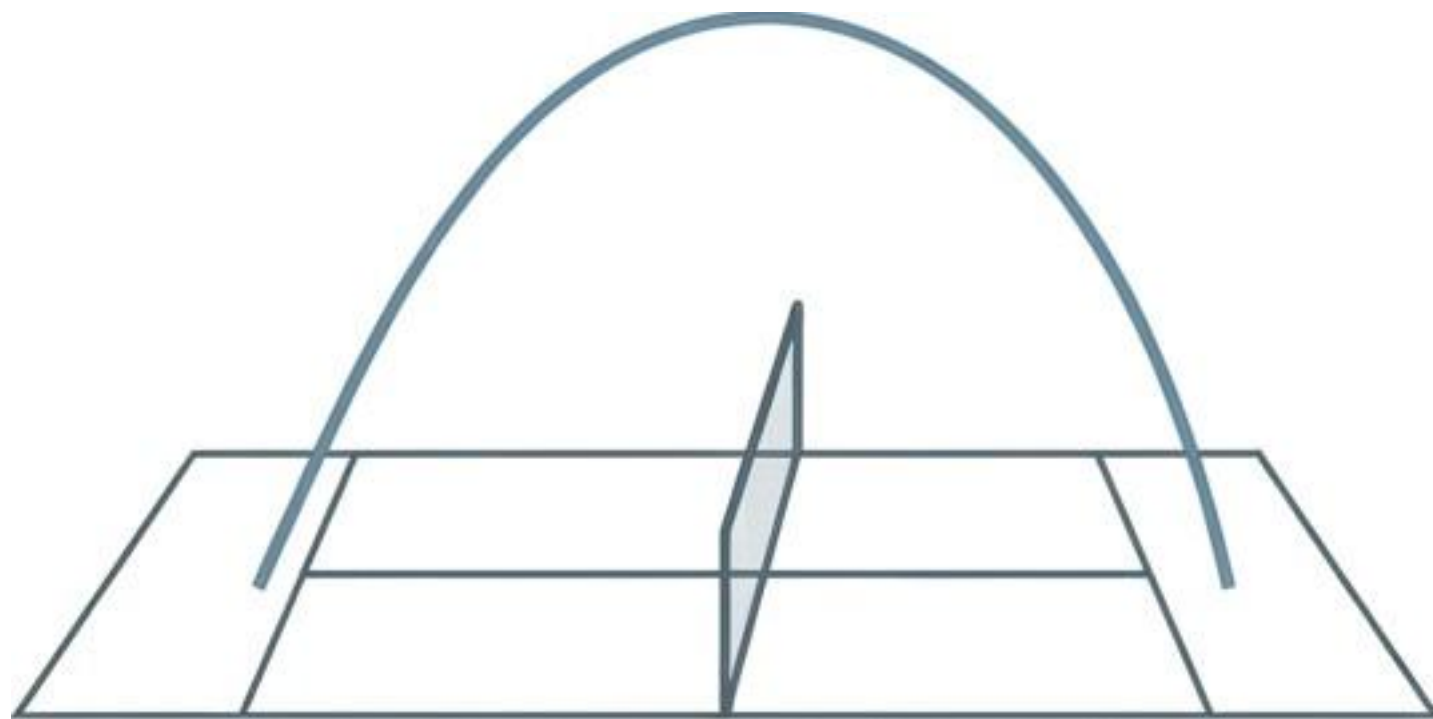
INCENTIVES (Dense Rewards):

- Making Racket Contact (+)
- Reducing Distance to Ball (+)
- Valid Ball Height (+)
- Valid Ball Height (+)

Phase 2: Competitive Self-Play



Inference Results: The Offense



Success Metrics:

- Motor Transfer: Effective use of Phase 1 skills.
- Trajectory: Generated shots that cleared the net and landed in bounds.
- Ball hit: Consistent approach and valid contact.

Verdict: Competent Ball Strikers.

Inference Results: The Defense



The Limitation: Reception Gap

- Stationary Failure: Agents often failed to move toward incoming shots.
- Game Dynamics: Short rallies. Exchanges dominated by Serve -> Return -> Point.
- Lack of Flow: Sustained back-and-forth play was rare.

Analysis: The Curriculum Gap



Diagnosis: The jump in complexity was too steep.

Effect: In self-play, agents converged on offensive strategies (fast wins) because defense was harder to learn.

The Path Forward



- 1. Intermediate Curriculum:
Insert a dedicated “Reception Training” phase.
- 2. 2v2 Expansion:
Introduce cooperative dynamics (Team Red vs Team Blue).

From individual motor skills to complex team strategy.’



Thank you!

